

two particles have the same amplitude.

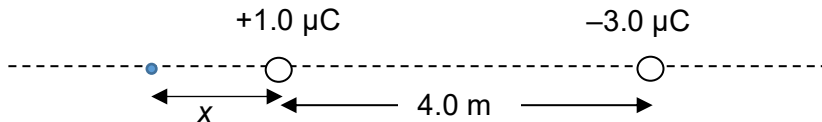
19

B

Location of neutral point cannot be between the two charges as they have opposite sign.

It must also be nearer the charge with smaller magnitude.

Thus, the neutral point is on the left of the  $+1.0 \mu\text{C}$  charge.



$$\text{Thus, } \frac{1}{4\pi\epsilon_0} \frac{1\mu}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{3\mu}{(x+4)^2}$$

$\Rightarrow x = 5.464 \text{ m}$  ; the negative root is rejected

20

A

Force on nuclei is in the same plane as the field lines. Since the nuclei are